Skills Summary

Dilations: Making and Measuring Similar Figures

Use this reference while completing your worksheet or when studying.

A dilation with centre C and scale factor k slides every point along the ray from C through it, until it stands k times as far from C. Angles never change and every length is multiplied by k, so the image is *similar* to the original — same shape, resized.

Dilating a point on the coordinate plane

Rule: Centred at the origin: $(x, y) \mapsto (kx, ky)$.

Centred at (h, j): $(x, y) \mapsto \big(h + k(x - h),\ j + k(y - j)\big)$ — measure from the centre, multiply by k, then step back out.

$k > 1$ enlarges, $0 < k < 1$ shrinks, and the centre itself never moves.

Example: Dilate $A(4, -6)$ about the origin with $k = 0.5$.
Step 1. The centre is the origin, so no shifting is needed — just multiply both coordinates by the scale factor.
Step 2. $0.5 \times 4 = 2$ and $0.5 \times (-6) = -3$.
$A' = \mathbf{(2,\ -3)}$

Try it: Dilate $B(-3, 10)$ about the origin with $k = 3$.

check: $\mathbf{(-9,\ 30)}$

Finding the scale factor from two lengths

Rule: $k = \dfrac{\text{image length}}{\text{matching pre-image length}}$ — image on top, always.

Any pair of corresponding sides gives the same k; if two pairs disagree, the figures are not similar and no dilation maps one to the other.

Example: A 12 cm segment is dilated onto an 18 cm segment. Find k.
Step 1. Scale factor compares image with pre-image, so divide — image on top, because the question asks what the original was multiplied by.
Step 2. $k = \dfrac{18}{12}$, and dividing both parts by 6 reduces it.
$k = \mathbf{\frac{3}{2}}$, an enlargement

Try it: A 20 mm segment is dilated onto an 8 mm segment. Find k.

check: $\mathbf{\frac{2}{5}}$

Using the scale factor on a similar figure

Rule: Under a dilation with factor k: every length (side, perimeter, diagonal, height) is multiplied by k; every angle is unchanged; every area is multiplied by k^2.

Perimeter is a length, so it takes k. Area is a product of two lengths, so it takes k twice.

Example: A triangle has sides 5, 6 and 9 cm and area 12 square cm. Dilate it with $k = 3$. Find the new perimeter and area.

Step 1. Perimeter is built from lengths, so it scales by k; area is built from two lengths, so it scales by k^2 — decide which before computing.

Step 2. Perimeter: $3 \times (5 + 6 + 9) = 3 \times 20$. Area: $12 \times 3^2 = 12 \times 9$.

Perimeter **60** cm, area **108** square cm

Try it: A rectangle has perimeter 24 cm and area 32 square cm. Dilate it with $k = 0.5$.

check: perimeter **12**, area **8**

(!) Watch out: Do not scale an area by k. A model at $k = \frac{1}{12}$ has $\frac{1}{144}$ of the original area, not $\frac{1}{12}$. And when the centre is not the origin, do not just multiply the coordinates — subtract the centre first, or the image lands in the wrong place.